Vivian W. H. Wong

Email: vivian.wong@ufl.edu
Website: vivian-wong.github.io
U.S. Permanent Resident

Assistant Professor, University of Florida
College of Design, Construction and Planning

Appointments and Research Experience

Academic

Assistant Professor, **University of Florida,** 2024 - Present
- College of Design, Construction and Planning, jointly appointed:
  - M.E. Rinker, Sr. School of Construction Management (50%)
  - Department of Urban and Regional Planning (50%)
- Affiliated with UF Artificial Intelligence and Informatics Research Institute (AIIRI)

Graduate Student Researcher, **Stanford University,** 2019 - 2024
- Stanford Center at the Incheon Global Campus (SCIGC)
- Engineering Informatics Group

Undergraduate Researcher, University of Illinois at Urbana-Champaign 2014 - 2017
- Smart Structures Technology Laboratory,
- Advisor: Billie F. Spencer, Jr.

Professional

Applied Scientist Intern, **Amazon Science**, Seattle, Summer 2022
- Topic: ML-enabled recommendation system, causal inference.
- Mentors: Michael Matheny, Michael D. Porter

Research Intern, **Alibaba**, Hangzhou, China Summer 2018
- Topic: Reinforcement learning for traffic signal control on MATSim
- Mentor: Liang Yu

Education

Stanford University, Stanford, CA 2024
Ph.D. in Civil Engineering, with minor in Computer Science
Advisor: Kincho H. Law
Dissertation title: Spatio-temporal Representation Learning: Applications to Manufacturing Planning and Pedestrian Crowd Analysis

Stanford University, Stanford, CA 2019
Master of Science in Civil Engineering (Structural Engineering & Mechanics)

University of Illinois at Urbana-Champaign, Champaign, IL 2017
Bachelor of Science in Civil Engineering

Research Areas

Theme: Sensing, AI, and simulation for adaptive decisions in physical and socio-technical systems, connecting observations to timely responses as conditions and resource availability change.

Smart manufacturing, operations, and logistics. Graph representations, reinforcement learning, and simulation for manufacturing dispatch, resource allocation, and post-disaster reconstruction.

Human mobility and accessible built environments. Multimodal sensing and machine learning for wheelchair accessibility assessment, pedestrian crowd forecasting, and understanding how physical conditions shape human movement.

Methods: Spatio-temporal learning, computer vision, graph neural networks, reinforcement learning, and discrete-event simulation.

Grants and Sponsored Research

Current Funded Research

G7 **University of Florida, Research Opportunity Seed Fund (Wong, PI)**
Total Award: $85,230
Project Period: 09/2025--08/2027
Project Title: Sensing for Empathetic Built Environments: Improving Wheelchair Accessibility and Inclusiveness on Campus through Data-Driven Analyses
Goal: Use wheelchair-mounted sensors and crowd measurements to identify barriers to campus mobility and inform improvements to accessibility.
Role: PI
Status: Awarded

Applications Pending

External Applications

G6 **National Science Foundation, Future CoRe: HCC (Li, PI)**
Total Award: $750,000 (requested)
Goal: Develop an expert-guided AI system that identifies and explains accessibility barriers in street-view images, using wheelchair measurements and community input to validate assessments and support planners.
Role: Co-PI
Status: Pending

G5 **Spencer Foundation, Vision Grants (Wu and Wong, PIs)**
Total Award: $75,000 (requested)
Goal: Bring together people with mobility impairments, campus staff, and researchers to connect accessibility evidence with coordinated institutional responses and plan a multi-university research program.
Role: Co-PI
Status: Pending

G4 **National Science Foundation, Presidential SI Challenge Supplemental Funding (Liu, PI)**
Total Award: $60,000 (requested)
Goal: Support high-school students in developing and testing AI-supported learning tools for interpreting building plans and exploring evacuation scenarios through faculty mentoring and project-based learning.
Role: Co-PI
Status: Pending

G3 **American Society of Interior Designers Foundation, Research Grant (Valipoor, PI)**
Total Award: $9,997 (requested)
Goal: Measure how spatial features and crowding affect wheelchair mobility in occupied buildings, using onboard sensors and participant feedback to develop evidence-based accessibility guidance.
Role: Co-I
Status: Pending

Internal Applications

G2 **University of Florida Claude D. Pepper Older Americans Independence Center, Pilot and Exploratory Studies Core (Zhang, PI)**
Total Award: $50,000 (requested)
Goal: Combine vehicle-mounted and wearable sensors with older adults' feedback to identify barriers to safe golf-cart use and guide transportation improvements that support mobility independence.

Role: Aim 1 Co-Lead (environmental sensing)
Status: Pending (letter of intent submitted)

Completed Funded Research

G1 **University of Florida, School of Landscape Architecture and Planning, Director's Interdisciplinary Innovation Grant (Luo, PI)**
Total Award: $9,978
Project Period: 08/2025--06/2026
Project Title: Explore Vision-Based AI's Usage in Landscape Performance Evaluation
Goal: Use video-based AI to measure how people use campus public spaces and how features such as seating and shade affect activity. Develop practical methods to support evidence-based design and management.
Role: Co-PI
Status: Completed

Research Supervision

Dongdong Wang, postdoctoral scholar.

Qianchen Yu, Ph.D. student. Reinforcement learning and simulation for post-disaster housing reconstruction; first author on [C7], [C9], and the manuscript under peer review [J5].

Ao He, Ph.D. student. Bicyclist safety and behavior using computer vision; first author of the cyclist-facility study accepted for presentation [C11] and co-author on crowd forecasting [C10].

Isabelle Gatmaitan, undergraduate researcher. Wheelchair accessibility sensing; first author of the BuildSys poster abstract [C8] and co-author on [C12].

Publications

** indicates a mentee/advisee.*

Journal Manuscripts under Peer Review

J5 Q. Yu*, Z. Zhang and **V. W. H. Wong**, ``A Sequential Decision Framework for Representing and Comparing Workforce Allocation Policies in Post-Disaster Building Recovery,'' under peer review at *Advanced Engineering Informatics*.
Corresponding author: Qianchen Yu.

Peer-Reviewed Journal Manuscripts

J4 M. Sato, **V. W. H. Wong**, H. Yeung, P. Witherell and K. H. Law, ``Identification and Interpretation of Melt Pool Shapes in Laser Powder Bed Fusion with Machine Learning'', *Smart and Sustainable Manufacturing Systems*, December 2024; 8(1): 1--23.
doi:10.1520/SSMS20230035

J3 **V. W. H. Wong**, S. H. Kim, J. Park, J. Park and K. H. Law, "Generating Dispatching Rules for the Interrupting Swap-Allowed Blocking Job Shop Problem Using Graph Neural Network and Reinforcement Learning", *ASME Journal of Manufacturing Science and Engineering*, Jan 2024; 146(1): 011009.
doi:10.1115/1.4063652

J2 **V. W. H. Wong** and K. H. Law, "Fusion of CCTV Video and Spatial Information for Automated Crowd Congestion Monitoring in Public Urban Spaces''. *Algorithms*, Mar 2023; 16(3):154.
doi:10.3390/a16030154

J1 **V. W. H. Wong**, M. Ferguson, K. H. Law, Y. T. Lee and P. Witherell, "Segmentation of Additive Manufacturing Defects Using U-Net''. *ASME Journal of Computing and Information Science in Engineering*, June 2022; 22(3):031005.
doi:10.1115/1.4053078

Conference Manuscripts under Peer Review

C14 **V. W. H. Wong**, ``Pickup and Placement Interleaving in Blocking Dual-Gripper Job Shops,'' *ASME International Manufacturing Science and Engineering Conference (MSEC 2027)*, Paper No. MSEC2027-203607.
Under peer review.

C13 A. He* and **V. W. H. Wong**, ``Video-Driven Agent-Based Simulation of Stadium Egress across Event Phases,'' under peer review at *IEEE International Conference on Big Data*.

Peer-Reviewed Conference Manuscripts (Published and Accepted)

C12 D. Wang*, A. Hagen, I. Gatmaitan*, H. Zhou, Y. Dong, S. Valipoor, **V. W. H. Wong**, L. Li, ``Do VLMs See What Sensors Feel? A Scalable Expert-Guided Design for Wheelchair Accessibility Assessment from Street View,'' *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027)*.
Accepted.
doi:10.48550/arXiv.2606.07642

C11 A. He*, M. Gupta, D. Salon, R. Eiris and **V. W. H. Wong**, ``Classifying Urban Locations by Cyclist Facility Usage and Directional Alignment: A Validated Camera-Based Framework from Multi-Site Observations,'' *106th Transportation Research Board (TRB) Annual Meeting*, Washington, D.C., Jan. 10--14, 2027. Accepted for presentation.
arXiv:2610.38045

C10 **V. W. H. Wong**, D. Wang*, A. He*, ``ST-DIF: A Spatio-Temporal Framework Integrating Adjacency Information in Crowd Forecasting,'' *ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys'26)*, Banff, AB, Canada, June 22-25, 2026.
Acceptance rate = 22.2%
doi:10.1145/3744256.3812585

C9 Q. Yu*, **V. W. H. Wong**, ``Resource-Constrained Scheduling for Post-Disaster Housing Reconstruction: A Reinforcement Learning-Based Simulation Study in Lombok, Indonesia,'' *Construction Research Congress (CRC)*, San Antonio, TX, Mar. 18-21, 2026.
doi:10.1061/9780784486986.064

C8 I. Gatmaitan*, Y. Dong, S. Valipoor, **V.W.H. Wong**, ``Poster Abstract: Dynamic Mobility Barrier Detection Using Wheelchair-mounted Sensors,'' *ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys'25)*, Golden, CO, Nov. 19-21, 2025, pp. 310-311.
doi:10.1145/3736425.3772111

C7 Q. Yu*, **V. W. H. Wong**, "PyRebuild: A Python-Based Simulator For the Dynamic Post-Earthquake Housing Reconstruction Problem,'' *ASCE International Conference on Computing in Civil Engineering (i3CE)*, New Orleans, LA, May 11-14, 2025.
doi:10.1061/9780784486443.034

C6 **V. W. H. Wong**, "Enhancing Data-Driven Predictive Modeling of Pedestrian Crowd Flow with Spatial Priors -- Case Studies with Post-Event Crowd Data on a University Campus," *2024 IEEE International Conference on Big Data (IEEE BigData)*, Washington, DC, Dec. 15-18, 2024, pp. 6795-6804.
Acceptance rate = 18.4%
https://doi.org/10.1109/BIGDATA62323.2024.10825781

C5 M. Sato, **V. W. H. Wong**, K. H. Law, H. Yeung and P. Witherell, "Explainability of Laser Powder Bed Fusion Melt Pool Classification Using Deep Learning'', *ASME Computers and Information in Engineering Conference (CIE)*. Aug. 20-23, 2023.
doi:10.1115/DETC2023-109137

C4 **V. W. H. Wong** and K. H. Law, "Modeling Crowd Data and Spatial Connectivity as Graphs for Crowd Flow Forecasting in Public Urban Space'', *ASCE International Conference on Computing in Civil Engineering (i3CE)*, Corvallis, OR, Jun. 25-28, 2023.
doi:10.1061/9780784485248.025

C3 M. Sato, **V. W. H. Wong**, K. H. Law, H. Yeung, Z. Yang, B. Lane and P. Witherell, ``Anomaly Detection of Laser Powder Bed Fusion Melt Pool Images'', *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (CIE)*, St. Louis, MO, Aug. 14-17, 2022.
doi:10.1115/DETC2022-88313

C2 **V. W. H. Wong**, M. Ferguson, K. H. Law, Y. T. Lee and P. Witherell, "Segmentation of Additive Manufacturing Defects Using U-Net", *ASME Computers and Information in Engineering Conference (CIE)*. Aug. 17-20, 2021.
doi:10.1115/DETC2021-68885

C1 **V. W. H. Wong**, M. Ferguson, K. H. Law and Y. T. Lee, "An Assistive Learning Workflow on Annotating Images for Object Detection", *2019 IEEE International Conference on Big Data*, Los Angeles, CA, USA, Dec. 9-12, 2019.
Acceptance rate = 18.7%
https://doi.org/10.1109/BIGDATA47090.2019.9006588

Peer-Reviewed Workshop Manuscripts

W1 R. Chen*, **V. W. H. Wong**, C. Grant, ``Measuring Decision-Scale Use in Tool-Augmented LLMs: A Contrastive Urban Benchmark,'' *AI Measurement Science (AIMS) Workshop at the Conference on Language Modeling (COLM 2026)*. Accepted (non-archival).
https://arxiv.org/abs/2609.16607

Preprints

P1 B. Lou, Z. Xu, **V. W. H. Wong**, ``CRoPE: Efficient Parametrization of Rotary Positional Embedding,'' *arXiv preprint* arXiv:2601.02728, 2026.
https://arxiv.org/abs/2601.02728

Teaching

Interests: Mechanics (statics, dynamics, mechanics of solids, vibrations, and finite element methods), grounded in structural-engineering and computational-mechanics training; programming and data structures (Python); AI theory and applications, including construction and the built environment; manufacturing and construction planning and scheduling; mechatronics (programmable electromechanical systems, digital and analog circuits, transistors, state machines, and C programming).

University of Florida: Instructor

Urban Data Structures (URP 6295), graduate — Fall 2026

AI Applications to Construction (BCN 5905), graduate — Fall 2026
Special Topics in Construction.

AI in the Built Environment (DCP 4300), undergraduate — Fall 2025
Mean instructor evaluation: 4.72/5.0; response rate above 60%. Recognized by UF Faculty Affairs for exceptional GatorEval scores.

Introduction to Planning Information Systems (URP 4273), undergraduate — Fall 2025
Mean instructor evaluation: 4.50/5.0.

University of Florida: Co-Instructor

Introduction to Planning Information Systems (URP 4273/6270) — Fall 2024
Undergraduate and graduate sections. Mean instructor evaluations: 4.61/5.0 (section 1, undergraduate) and 4.57/5.0 (section 2, graduate). Recognized by UF Faculty Affairs for exceptional GatorEval scores.

Stanford University: Graduate Teaching Assistant

Optimization in Structural Engineering (CEE 380) — Spring 2023

Finite Element Methods in Structural Dynamics (CEE 284) — Autumn 2019, 2021, 2022
Structural Dynamics (CEE 283) — Winter 2020, 2022

University of Illinois at Urbana-Champaign

Engineering Orientation (ENG 100), co-instructor — Fall 2016
Engineering First-Year Experience Seminars (ENG 177), teaching assistant — Fall 2016

Professional Service

Conferences

ACM BuildSys'25 Publication Chair — 2025

Paper Reviewer

Engineering Applications of Artificial Intelligence — 2025
IEEE Transactions on Neural Networks and Learning Systems — 2023
Optimization Letters — 2022
Automation in Construction — 2022
IEEE Big Data Conference — 2019, 2026

Industry Engagement

Involved with the Rinker School Industry Advisory Board (RSIAB) — 2024 - Present

Member Affiliations

ACM
ASCE
IEEE
Society of Women Engineers (SWE)
Women in Science and Engineering (WISE) Group
Someone Like Me (SLM) Mentorship Program
Women's Community Center (WCC) STEM Mentorship Program

Leadership and Broadening Participation

Faculty Advisor — 2024 - Present
University of Florida Table Tennis Club

Student Athlete, Stanford Table Tennis Team — 2022 - 2024

Mentor — 2023
Stanford Women Community Center (WCC) STEM mentorship program
Mentees: Lauren Williams, Jayna Huang

Coordinator — 2023
Stanford Someone Like Me (SLM) mentorship program

Residence Community Associate — 2018-2019
Stanford Kennedy Graduate Residence / Graduate Life Office

Always Connecting Representative — 2017
Society of Women Engineers annual conference [link to media coverage]

Volunteer Instructor — 2017
Shakes and Quakes outreach at Sangamon Elementary School

Co-Founder, President and Project Manager — 2016 - 2017
Engineers In Action (EIA) UIUC for suspension bridge construction in underdeveloped communities in Guatemala and Panama

Field Representative — 2016
Saha Global for delivering electricity and entrepreneurial opportunities for women in Ghana

Officer and Nominating Committee 2015 - 2017
Society of Women Engineers at UIUC

Awards and Honors

Blume Fellowship, Stanford University	2019
James Scholar, UIUC	2014 - 2017
Earle J. Wheeler Scholarship, UIUC	2016
Fred S. Bailey International Service Scholarship for Cause-Driven Leaders, UIUC	2016
International Engineering Fellowship, UIUC	2016
Wayne C. Teng Scholarship, UIUC	2015

Invited Talks

Modeling and Learning Manufacturing and Facility Management: A Spatio-Temporal Graph Approach 05/2026
Guest speaker, Montgomery Blair High School, Silver Spring, MD

From Static to Responsive: Utilizing Multimodal Data and ML to Understand Dynamic Human Behaviors 02/2026
Transportation Seminar, University of South Florida, Tampa, FL

ST-DIF Python Package: Spatio-temporal Data Integration System for Pedestrian Crowd Flow 07/2025
Nigiwai: Placemaking Driven by Human Behavior, Stanford, CA

Spatiotemporal Data to Understand Human Behavior and Mobility in Urban Systems 05/2023
TU Delft--Stanford: Designing for Future Mobility Workshop, Stanford, CA

Understanding Human Behaviors in Smart Building and Urban Environments 06/2021
Stanford Center at the Incheon Global Campus First International Symposium, Online